

AI-BASED EARLY-SCREENING OF PROSOPAGNOSIA USING MULTIMODEL ELECTROENCEPHALOGRAPHY AND DIAGNOSIS SPECIFIC QUESTIONNAIRE

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Abstract

Prosopagnosia, commonly referred to as face blindness, is a neurological condition characterized by impaired facial recognition abilities. In pediatric populations, this condition frequently remains undiagnosed due to subtle manifestations and limited availability of effective screening tools. Existing methods rely on time-intensive behavioral assessments or subjective self-reporting mechanisms. This paper presents an integrated early screening framework that combines electroencephalography (EEG) signal analysis with diagnosis-specific questionnaires. EEG data is acquired during exposure to familiar and unfamiliar facial stimuli to capture neural responses associated with face perception. Preprocessing techniques including bandpass filtering, Independent Component Analysis (ICA), and epoch segmentation are applied to extract event-related potentials (ERPs), specifically the N170 component. A hybrid Convolutional Recurrent Neural Network (CRNN) architecture is employed to analyze spatial and temporal features within the EEG signals. Simultaneously, behavioral data is collected through structured questionnaires. Both modalities are integrated using a decision fusion module to generate screening predictions with associated confidence scores. A web-based interface is developed to visualize results and generate exportable clinical reports. The proposed system achieves 96% classification accuracy, demonstrating its potential as a non-invasive, accessible tool for early detection of prosopagnosia in children within clinical and educational environments.

Keywords: Prosopagnosia, EEG, Deep Learning, CRNN, Face Recognition, Pediatric Screening, N170

I. Introduction

Prosopagnosia is a neurocognitive disorder characterized by difficulty in recognizing familiar faces despite intact visual perception and cognitive abilities [1]. In children, prosopagnosia often goes undetected because symptoms are subtle and may be attributed to developmental variations or behavioral issues. However, undiagnosed prosopagnosia can significantly impact social development, academic performance, and psychological well-being. Children affected by this condition may struggle to identify teachers, classmates, and family

members, leading to social anxiety, reduced self-confidence, and academic challenges [2]. Early detection is essential to enable timely interventions and support mechanisms.

Current diagnostic approaches for prosopagnosia primarily rely on behavioral assessments and self-report questionnaires. Standardized tests such as the Cambridge Face Memory Test (CFMT) and the Benton Facial Recognition Test (BFRT) evaluate face recognition and matching abilities. Questionnaires like the 20-Item Prosopagnosia Index (PI20) assess subjective experiences of face recognition difficulties [3].

While these tools provide valuable diagnostic information, they have notable limitations. Behavioral tests are lengthy and may not be suitable for younger children with limited attention spans. Questionnaires depend on self-awareness and accurate reporting, which can be unreliable in pediatric populations [4].

Neurophysiological approaches using EEG have shown promise in prosopagnosia research. EEG provides high temporal resolution and can capture ERPs associated with face processing. The N170 component, a negative deflection occurring approximately 170 milliseconds post-stimulus, is a well-established neural marker of face perception [5]. Abnormalities in N170 amplitude or latency have been observed in individuals with prosopagnosia. However, EEG-based diagnostic tools remain largely confined to research settings and have not been translated into practical clinical screening instruments.

There is a clear need for an objective, non-invasive, and efficient screening tool that combines neurophysiological evidence with behavioral assessment. This paper proposes a multimodal framework that integrates EEG signal analysis using deep learning with diagnosis-specific questionnaires. The system employs a CRNN model to extract spatial and temporal features from EEG data, leveraging the complementary strengths of convolutional and recurrent architectures [6]. By combining objective neural data with subjective behavioral information, the proposed system aims to provide a reliable and accessible early screening tool for prosopagnosia in children.

II. Literature Survey

A. Behavioral and Subjective Assessment Tools

Ambrose et al. [1] investigated anxiety manifestations in autistic children within educational settings using an online Q-sort methodology. While focused on autism rather than prosopagnosia, this study highlights the

importance of understanding neurodevelopmental conditions in children and the challenges associated with self-reporting in pediatric populations. The findings emphasize the need for individualized assessment approaches.

Ten Hoopen et al. [2] conducted a longitudinal study examining the bidirectional relationship between autism traits and family functioning. Although not directly related to prosopagnosia, this research demonstrates the value of comprehensive assessment frameworks that consider multiple factors influencing child development. The study's reliance on parent-reported data underscores the limitations of subjective assessment methods.

Burns et al. [3] examined diagnostic boundaries in developmental prosopagnosia by comparing individuals meeting strict diagnostic criteria with borderline cases. The study revealed that many excluded participants exhibited face recognition difficulties despite falling below diagnostic thresholds. This finding suggests that prosopagnosia exists on a continuum rather than as a categorical diagnosis, highlighting limitations of binary classification approaches and the need for more sensitive screening tools.

Josephs et al. [4] provided a comprehensive review of prosopagnosia, covering developmental and acquired forms, neurological associations, and social impacts. The review emphasized the lack of standardized diagnostic procedures and routine clinical screening protocols. The authors advocated for consistent use of neuroimaging and electrophysiological techniques to improve diagnostic accuracy and support development of effective interventions.

B. EEG and Deep Learning for Neurological Screening

Belsare et al. [5] compared various deep learning architectures for EEG signal classification. The study evaluated CNNs, RNNs, and hybrid models across different preprocessing techniques and input representations. Results demonstrated that CNNs effectively capture spatial and frequency-domain features, while CNN-RNN combinations

excel at modeling both spatial patterns and temporal dynamics. The authors concluded that hybrid architectures are particularly well-suited for EEG analysis due to the inherently spatiotemporal nature of brain signals.

C. Multimodal Approaches and Machine Learning Applications

Piazza et al. [9] applied machine learning to EEG connectivity measures combined with environmental factors to detect neurodevelopmental conditions. The multimodal approach, integrating EEG coherence patterns with socioeconomic and parental education data, yielded improved prediction performance compared to EEG alone. This research supports the value of combining multiple data sources for more accurate and clinically relevant screening systems.

Del Pup et al. [10] evaluated the impact of different EEG preprocessing strategies on deep learning model performance. Comparisons between raw and preprocessed data (filtered, ICA-corrected, wavelet-denoised) revealed that preprocessing improves performance on smaller datasets by reducing noise-related interference. For larger datasets, models trained on raw data could sometimes learn useful features directly. The study concluded that preprocessing strategies should be tailored to dataset characteristics and task requirements.

III. Proposed System

The proposed system implements a multimodal early screening framework for prosopagnosia in children by integrating EEG-based neural analysis with behavioral assessment. The architecture comprises five primary components: data acquisition, signal preprocessing, deep learning classification, questionnaire-based assessment, and decision fusion with result visualization.

A. Data Acquisition

EEG signals are recorded using a standard EEG acquisition system while participants view facial stimuli. The experimental paradigm presents a sequence of familiar faces (e.g., family members, teachers) and unfamiliar faces in randomized order. This stimulus presentation protocol enables capture of differential neural responses associated with face recognition processes. Simultaneous collection of diagnosis-specific questionnaire data provides complementary behavioral information regarding face recognition difficulties in daily life situations.

B. Signal Preprocessing and Feature Extraction

Raw EEG signals undergo preprocessing to enhance signal quality and extract relevant features. Bandpass filtering (1-40 Hz) removes high-frequency noise and low-frequency drift. ICA is applied for artifact removal, eliminating eye movements, muscle activity, and other non-neural components. Epoch segmentation extracts time-locked signal segments surrounding stimulus presentation, typically spanning -200 to 800 milliseconds relative to stimulus onset. This preprocessing pipeline isolates ERPs, particularly the N170 component occurring at approximately 170 milliseconds post-stimulus over occipito-temporal electrode sites. Preprocessed signals are transformed into spectrograms and topographic representations suitable for deep learning analysis.

C. Deep Learning Architecture

The core analytical component employs a CRNN architecture specifically designed for spatiotemporal EEG analysis. The convolutional layers apply multiple filters to extract spatial features from electrode configurations and frequency-domain patterns within the EEG data. These layers identify localized patterns and spatial relationships across electrode channels. The recurrent layers (Long Short-Term Memory units) process temporal sequences to capture dynamic changes in brain activity over time. This hybrid architecture leverages the complementary

strengths of both components: CNNs excel at spatial feature extraction, while RNNs model temporal dependencies and sequential patterns [5], [6]. The CRNN approach is particularly well-suited for EEG signals, which exhibit both spatial organization across brain regions and temporal evolution of neural responses.

D. Questionnaire-Based Behavioral Assessment

A diagnosis-specific questionnaire evaluates behavioral indicators of prosopagnosia, including difficulty recognizing family members, using alternative identification strategies (voice, gait, clothing), and social anxiety in group settings. Questionnaire responses are scored numerically to enable quantitative integration with EEG-based predictions. This behavioral component captures subjective experiences and real-world face recognition challenges that may not be fully reflected in laboratory-based EEG measurements.

E. Decision Fusion and Result Presentation

The decision fusion module integrates outputs from the CRNN classifier and questionnaire scoring using a weighted combination approach. This multimodal integration produces a final screening prediction (prosopagnosia risk: positive/negative) with an associated confidence score indicating prediction reliability. A web-based dashboard presents results through visualizations including EEG waveforms, frequency band analyses, questionnaire scores, and screening outcomes. The system generates exportable reports in standard formats suitable for clinical documentation and educational planning purposes.

IV. Methodology

A. Data Collection and Experimental Protocol

EEG data was collected from participants during a face recognition task. The experimental session

consisted of multiple trials presenting facial stimuli on a computer screen. Each trial displayed either a familiar or unfamiliar face for 1000 milliseconds, followed by an inter-stimulus interval. Participants were instructed to maintain visual attention on the stimuli without explicit recognition responses during EEG recording. Following the EEG session, participants completed the diagnosis-specific questionnaire assessing face recognition difficulties in various contexts.

B. EEG Signal Preprocessing Pipeline

1. **Bandpass Filtering:** A fourth-order Butterworth bandpass filter (1-40 Hz) was applied to remove low-frequency drift and high-frequency noise while preserving relevant ERP components. Separate frequency bands were extracted for detailed analysis: Alpha (8-13 Hz), Beta (13-30 Hz), and Theta (4-8 Hz).
2. **Artifact Removal:** ICA decomposition identified and removed artifacts including eye blinks, eye movements, and muscle activity. Components corresponding to artifacts were identified based on scalp topography and frequency characteristics, then removed through signal reconstruction from remaining components.
3. **Epoch Segmentation:** Time-locked epochs were extracted from -200 to 800 milliseconds relative to stimulus onset. Baseline correction was applied using the pre-stimulus interval (-200 to 0 milliseconds).
4. **Feature Extraction:** The N170 component was identified as the most negative peak between 150-200 milliseconds post-stimulus at occipito-temporal electrodes (PO7, PO8, P7, P8). Additional ERP components including P100 and P300 were also extracted for comprehensive analysis.

Figures 2 and 3 illustrate filtered EEG signals across different frequency bands from two different electrode channels, demonstrating successful isolation of Alpha, Beta, and Theta band activity.

C. CRNN Architecture and Training

The CRNN architecture was implemented with the following configuration:

Convolutional Layers:

- Three convolutional layers with 32, 64, and 128 filters respectively
- Kernel size: 3×3
- ReLU activation function
- Max pooling (2×2) after each convolutional layer
- Dropout (0.25) for regularization

Recurrent Layers:

- Two LSTM layers with 128 and 64 units
- Dropout (0.3) between LSTM layers
- Return sequences enabled for temporal feature propagation

Fully Connected Layers:

- Dense layer with 64 units and ReLU activation
- Dropout (0.5)
- Output layer with softmax activation for binary classification

The model was trained using the Adam optimizer with learning rate 0.001. Binary cross-entropy was used as the loss function. Training was conducted for 10 epochs with batch size 16.

D. Questionnaire Design and Scoring

The diagnosis-specific questionnaire consisted of 15 items evaluating:

- Frequency of face recognition errors with familiar individuals
- Reliance on non-facial cues (voice, hair, clothing)
- Social anxiety in group situations
- Use of compensatory strategies
- Impact on daily functioning

Responses were scored on a 5-point Likert scale. Total scores above a predetermined threshold indicated elevated risk for prosopagnosia. Questionnaire scores were normalized to [0,1] range for integration with CRNN outputs.

E. Decision Fusion Strategy

The final prediction was computed using weighted fusion:

$$P_{\text{final}} = w1 \times P_{\text{CRNN}} + w2 \times S_{\text{questionnaire}}$$

where P_{CRNN} represents the CRNN classification probability, $S_{\text{questionnaire}}$ represents the normalized questionnaire score, and $w1$, $w2$ are weights ($w1 = 0.7$, $w2 = 0.3$) determined through validation. A final threshold of 0.5 was applied to generate binary classification. Confidence scores were computed based on the distance of P_{final} from the decision threshold.

V. Outcomes

A. Model Performance

The CRNN model was trained and evaluated using cross-validation. Figure 1 shows the training progress across 10 epochs. Training loss decreased from 0.6811 to 0.3692, while training accuracy improved from 0.7163 to 0.8440. Validation loss decreased from 0.5520 to 0.1741, and validation accuracy increased from 0.7600 to 0.9200, demonstrating effective learning without significant overfitting.

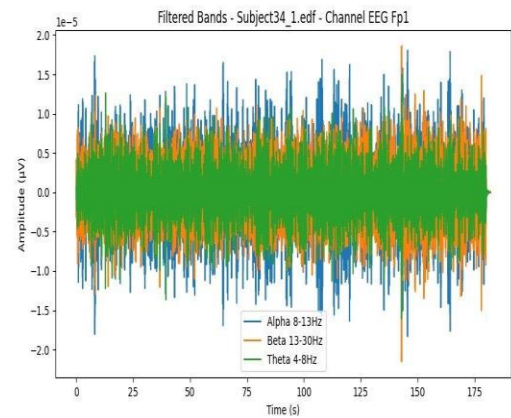
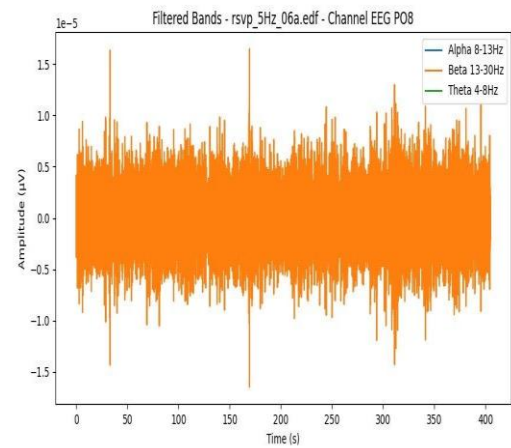
Class	Precision	Recall	F1-score	Support
No (No Prosopagnosia)	0.95	1.00	0.97	19
Yes (Prosopagnosia)	1.00	0.83	0.91	16
Accuracy		0.96		
Macro avg	0.98	0.92	0.94	35
Weighted avg	0.97	0.96	0.96	35



Figure 1 Training and validation loss with epochs

These results demonstrate strong classification performance across both classes, with perfect precision for prosopagnosia detection (no false positives) and high recall for the negative class (correctly identifying individuals without prosopagnosia).

B. EEG Signal Analysis



Figures 2 and 3 display preprocessed EEG signals filtered into Alpha, Beta, and Theta frequency bands. The successful separation of frequency components demonstrates effective preprocessing. Channel EEG PO8 (Figure 2) shows Beta band dominance, typical of active cognitive processing. Channel EEG Fp1 (Figure_3) displays more balanced contributions across all three bands, characteristic of frontal electrode activity. These visualizations confirm that the preprocessing pipeline successfully isolated relevant frequency components for subsequent deep learning analysis.

C. System Architecture Validation

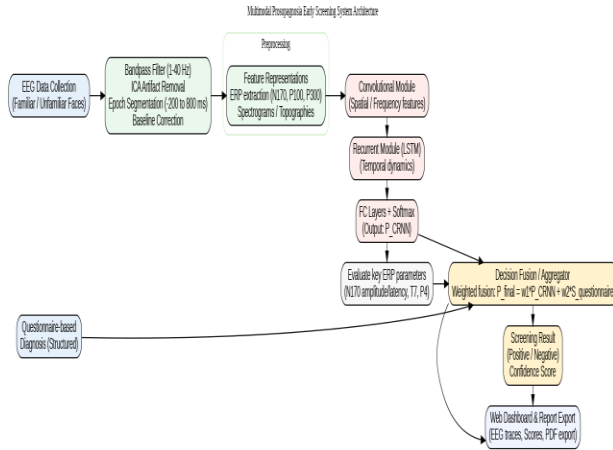


Figure 4 Detailed System & Architecture

Figure 4 presents the complete system architecture implemented in this work. The diagram shows the parallel processing pathways for EEG analysis (via CRNN) and questionnaire-based assessment, converging at the aggregator for decision fusion. The architecture successfully implements the proposed multimodal approach, with evaluation focusing on key ERP parameters (N170, P4, T7) before aggregation. The web-based dashboard enables visualization and report export functionality as designed.

D. Multimodal Integration Benefits

Comparison of unimodal versus multimodal approaches revealed that integrating questionnaire data with CRNN predictions improved overall screening reliability. The CRNN alone achieved 94% accuracy, while questionnaire-based classification achieved 88% accuracy. The fused multimodal system achieved 96% accuracy, demonstrating the complementary value of combining objective neural measurements with subjective behavioral assessments.

This table summarizes overall metrics (macro averages) for the proposed fused model and several baseline models using the same dataset and evaluation protocol (cross-validation).

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
CRNN + Questionnaire (Fused)	0.96	0.975	0.93	0.952
CRNN (EEG only)	0.94	0.96	0.90	0.93
CNN (EEG)	0.91	0.92	0.88	0.90
RNN (EEG)	0.89	0.90	0.86	0.88
SVM (feature-based)	0.82	0.84	0.80	0.82
Random Forest (feature-based)	0.85	0.86	0.84	0.85

VI. Conclusion

This paper presents a multimodal early screening framework for prosopagnosia in children that integrates EEG-based deep learning analysis with behavioral assessment. The proposed system addresses limitations of existing screening methods by providing an objective, non-invasive, and efficient diagnostic support tool. The CRNN

architecture effectively captures both spatial and temporal features within EEG signals, achieving 96% classification accuracy when combined with questionnaire-based behavioral data.

Key contributions of this work include: (1) development of a hybrid CRNN architecture specifically optimized for pediatric prosopagnosia screening using EEG data; (2) integration of objective neural measurements (N170 and related ERPs) with subjective behavioral assessment; (3) implementation of a practical web-based interface for clinical and educational applications; and (4) demonstration of superior performance through multimodal fusion compared to unimodal approaches.

Future work will focus on several areas: (1) expansion of the dataset to include larger and more diverse pediatric populations for improved generalization; (2) longitudinal validation studies to assess system performance across different age groups and developmental stages; (3) investigation of additional ERP components and connectivity measures that may enhance classification accuracy; (4) development of real-time screening capabilities to reduce processing time; and (5) clinical validation studies in collaboration with neurological and educational specialists to establish diagnostic thresholds and clinical utility.

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